

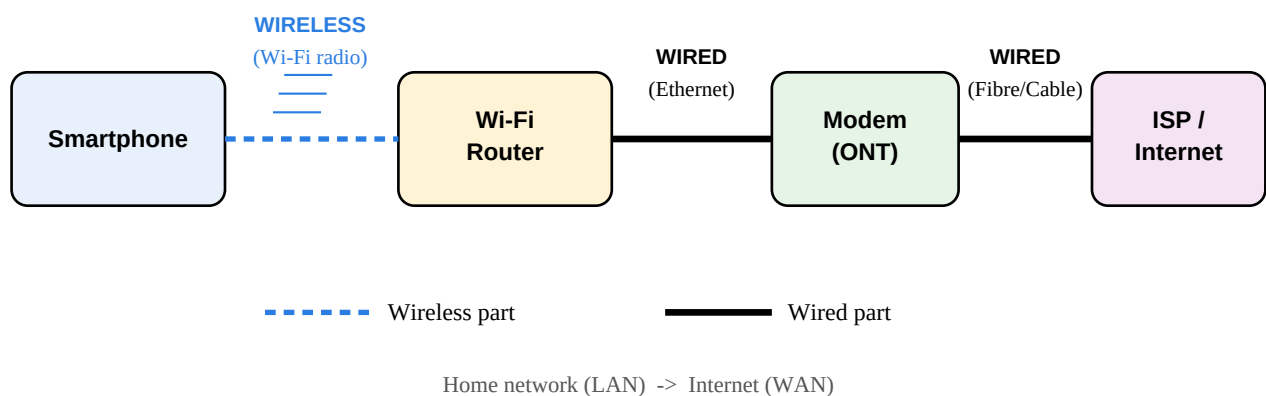
Wireless Networking

Answers to Questions 1-4

1. Three popular Wi-Fi standards

Standard (Wi-Fi name)	Frequency band	Key difference
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Introduced MIMO; speeds up to about 600 Mbps. Works on both bands, so it is widely compatible but slower than newer standards.
802.11ac (Wi-Fi 5)	5 GHz only	Much faster than 802.11n (theoretical several Gbps) thanks to wider channels (up to 160 MHz) and MU-MIMO on the 5 GHz band.
802.11ax (Wi-Fi 6/6E)	2.4 GHz, 5 GHz (6 GHz in Wi-Fi 6E)	Better efficiency in crowded places using OFDMA, giving higher real-world speed with many devices; Wi-Fi 6E adds the less congested 6 GHz band.

2. Diagram: smartphone to internet via home Wi-Fi router



The phone talks to the router over radio waves (wireless). From the router onward, data travels over cables (wired) through the modem to the ISP and then the internet.

3. Scanning Wi-Fi networks

Note: I cannot scan networks from here, so the table below uses **sample values**. Replace them with what your own phone or laptop shows (Wi-Fi settings, or the apps WiFi Analyzer / inSSIDer).

SSID	Signal strength	Security type
Home_WiFi_5G	-45 dBm (Excellent)	WPA3
Neighbor_Net	-60 dBm (Good)	WPA2
Free_Cafe_WiFi	-50 dBm (Very good)	Open (no security)

My choice: Home_WiFi_5G. It has the strongest signal (closest to 0 dBm is better) and the strongest security (WPA3). Free_Cafe_WiFi has a good signal, but an open network has no encryption, so others can intercept my traffic. Neighbor_Net uses WPA2, which is decent, but its signal is weaker

and I do not own or control that network. A good choice needs both a strong signal and strong security.

4. SSID hiding

What it is: Every Wi-Fi router normally broadcasts its network name (SSID) in beacon frames so nearby devices can see it in the list of networks. With SSID hiding, the router stops broadcasting the name, so the network does not appear in the normal list. To connect, a user must type the exact SSID and password manually.

Advantage: The network is less visible to casual users and neighbours, which reduces random connection attempts and makes it look less like an easy target.

Disadvantage: It is not real security. The SSID is still present in other frames (for example when a device connects), so tools like Wireshark can reveal it easily. It also makes setup less convenient, and devices must keep probing for the hidden name, which can reveal it in other places too. Strong encryption (WPA2/WPA3) with a good password is the proper protection.